

29. Two infinitely long parallel conductors A and B, separated by 2 cm, carry 6 A and 2 A currents in opposite directions respectively. Find :

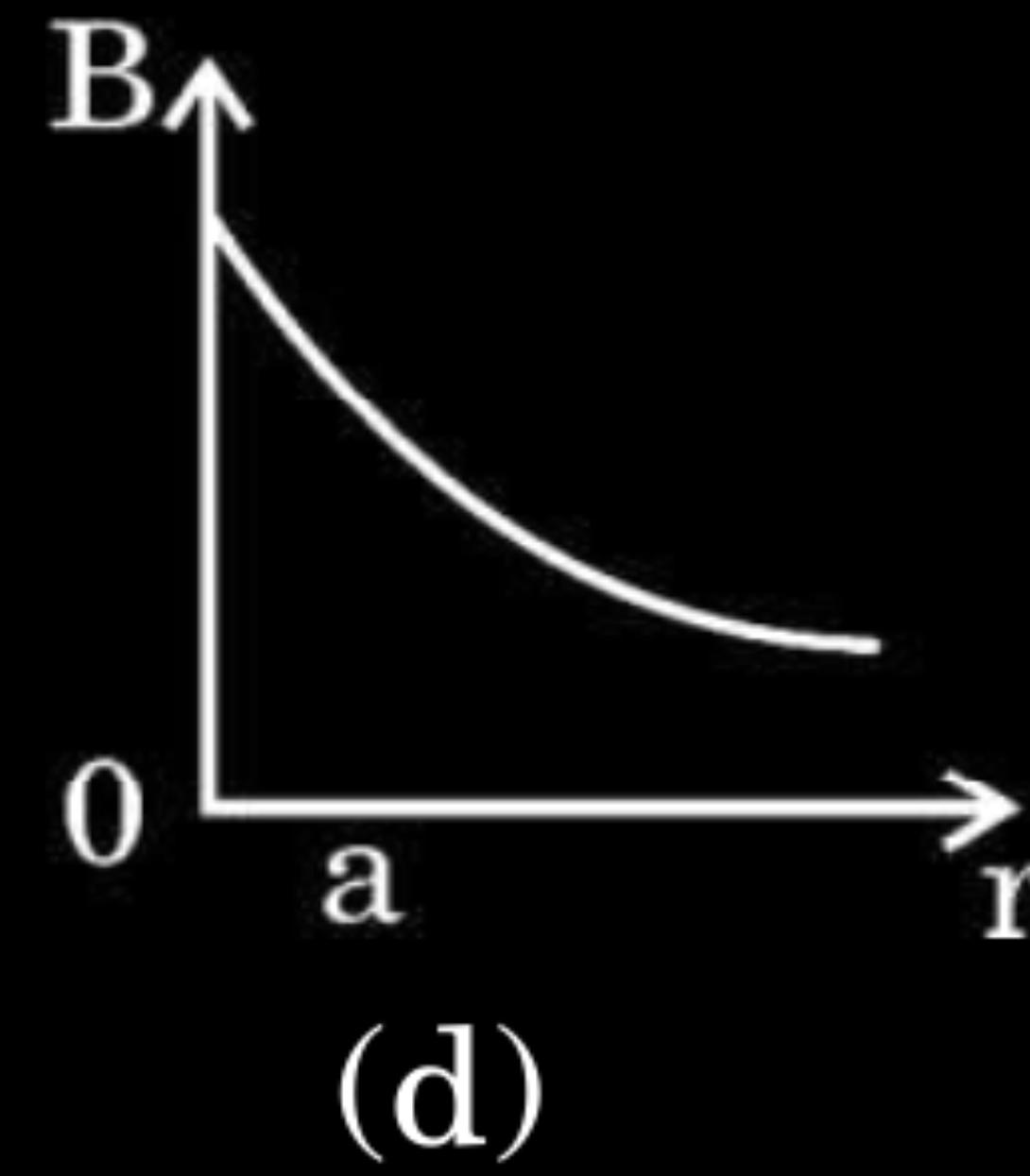
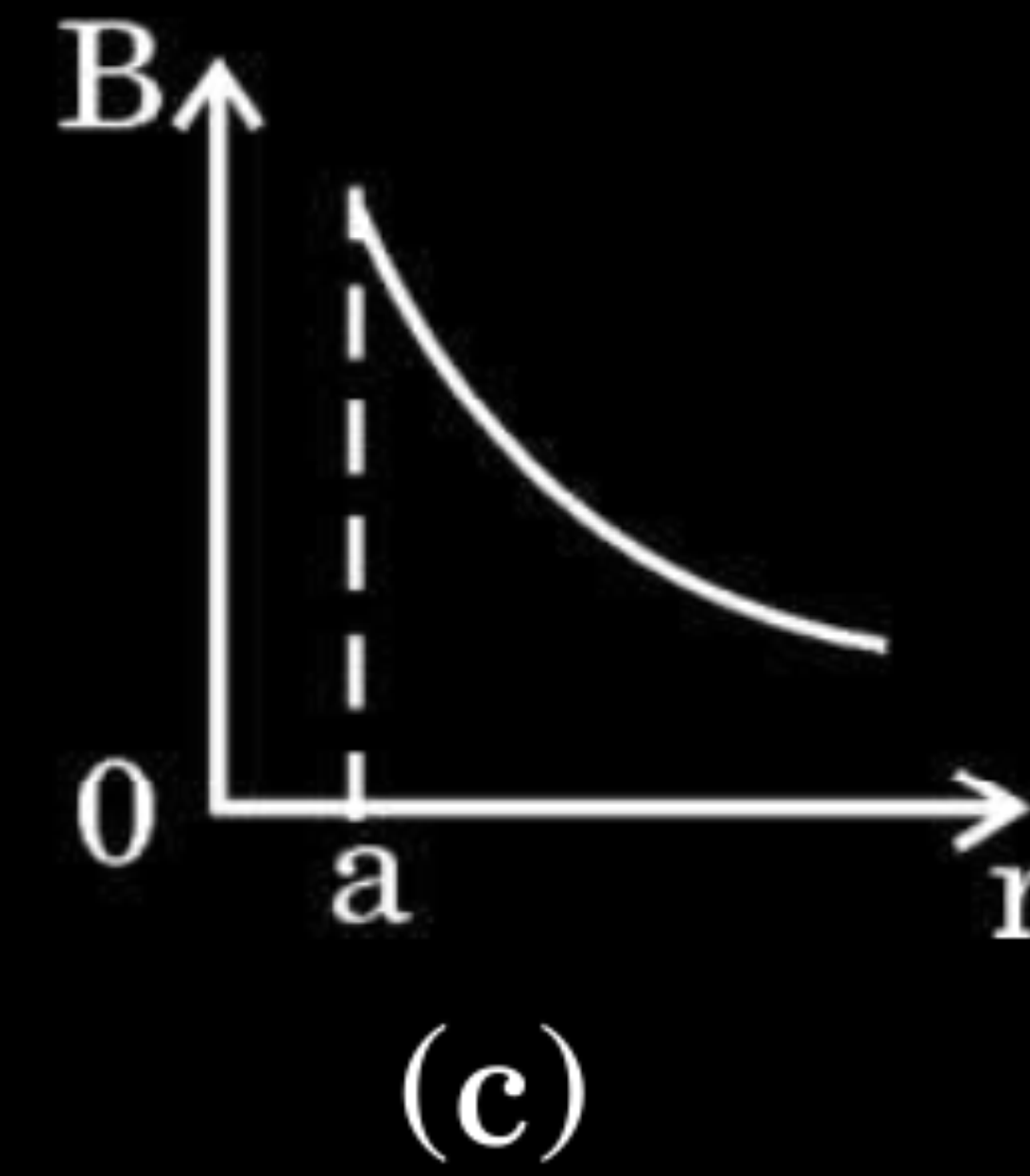
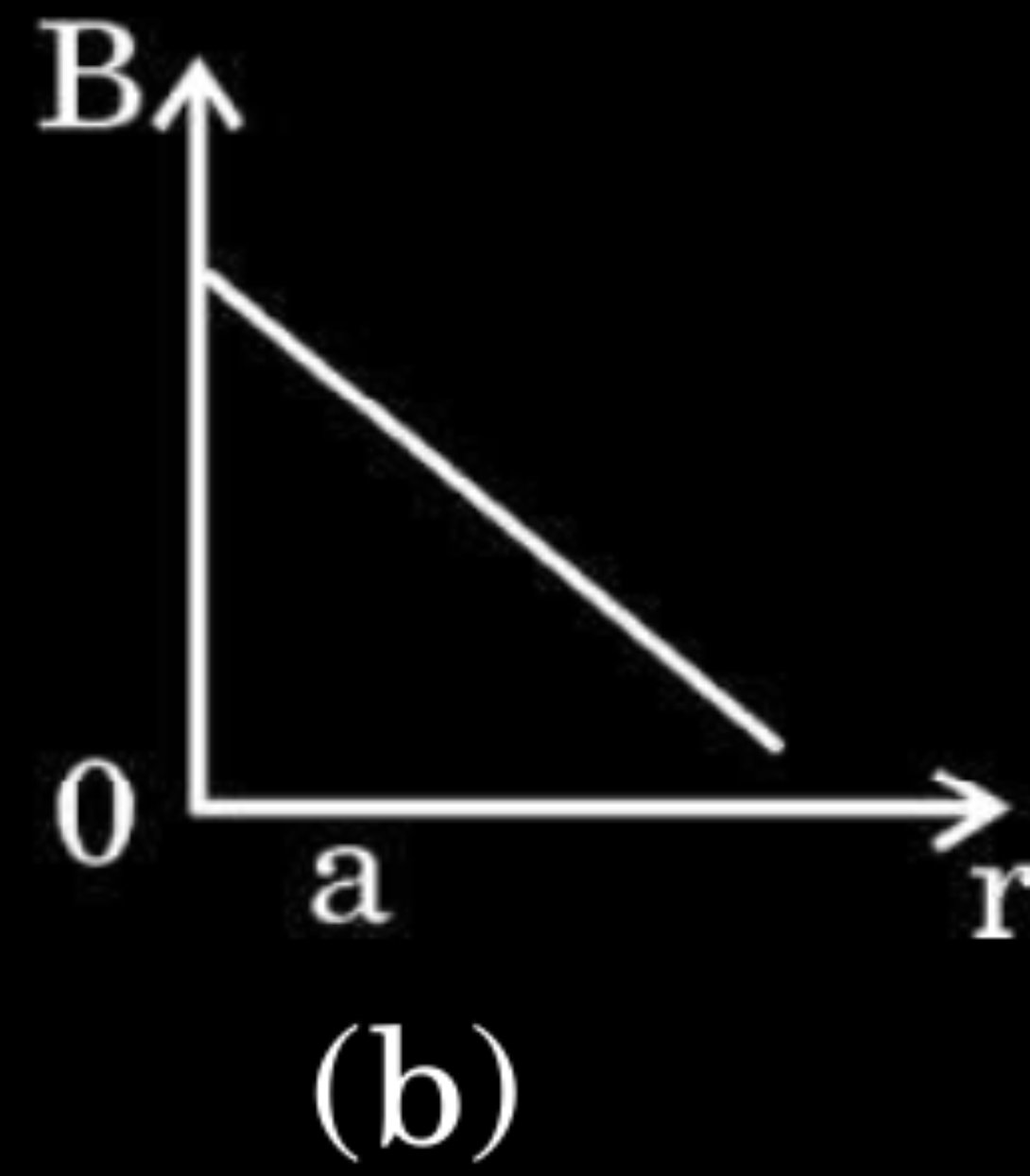
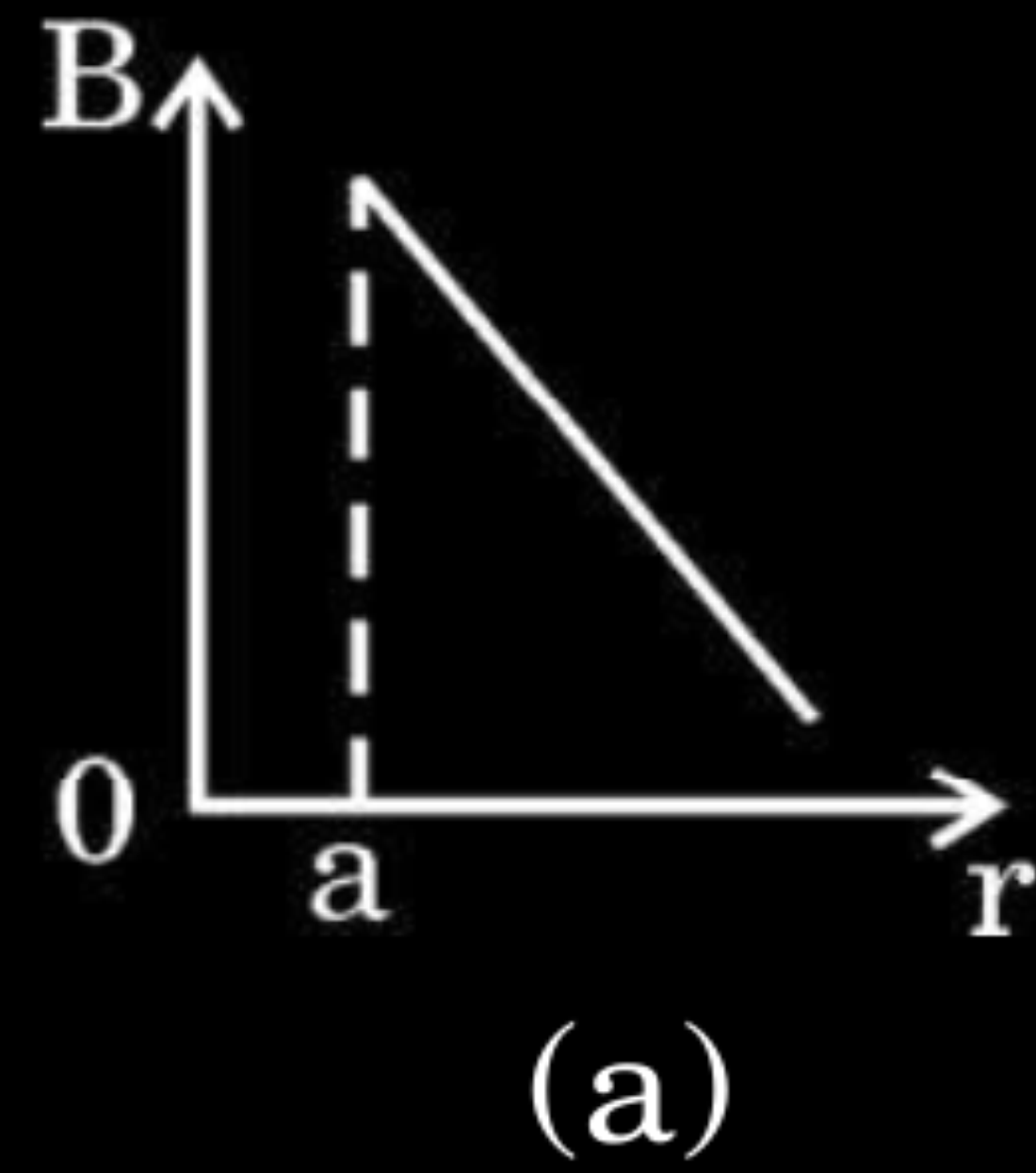
- (a) the net magnetic field at a point midway between A and B.
- (b) the force acting per unit length on B.

3

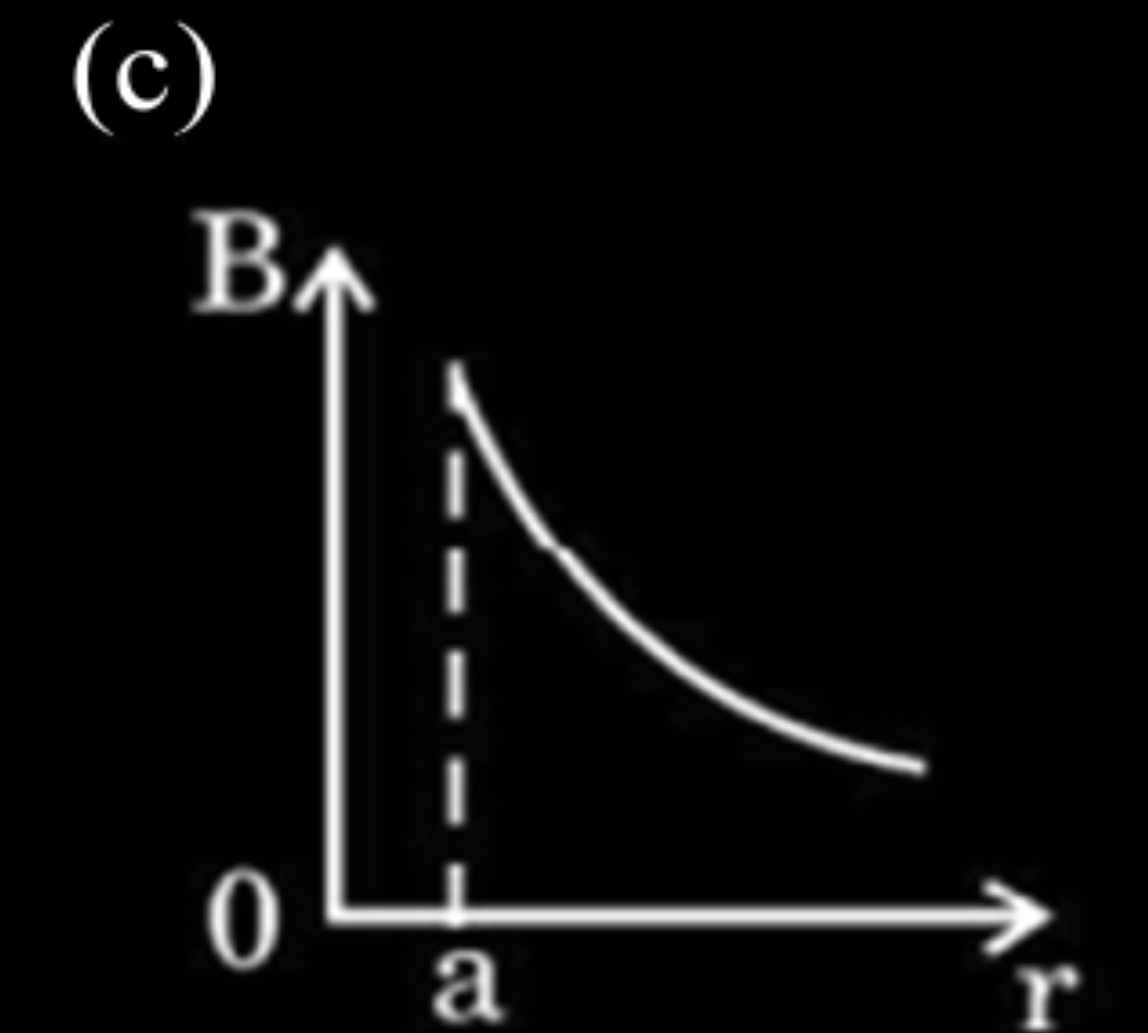
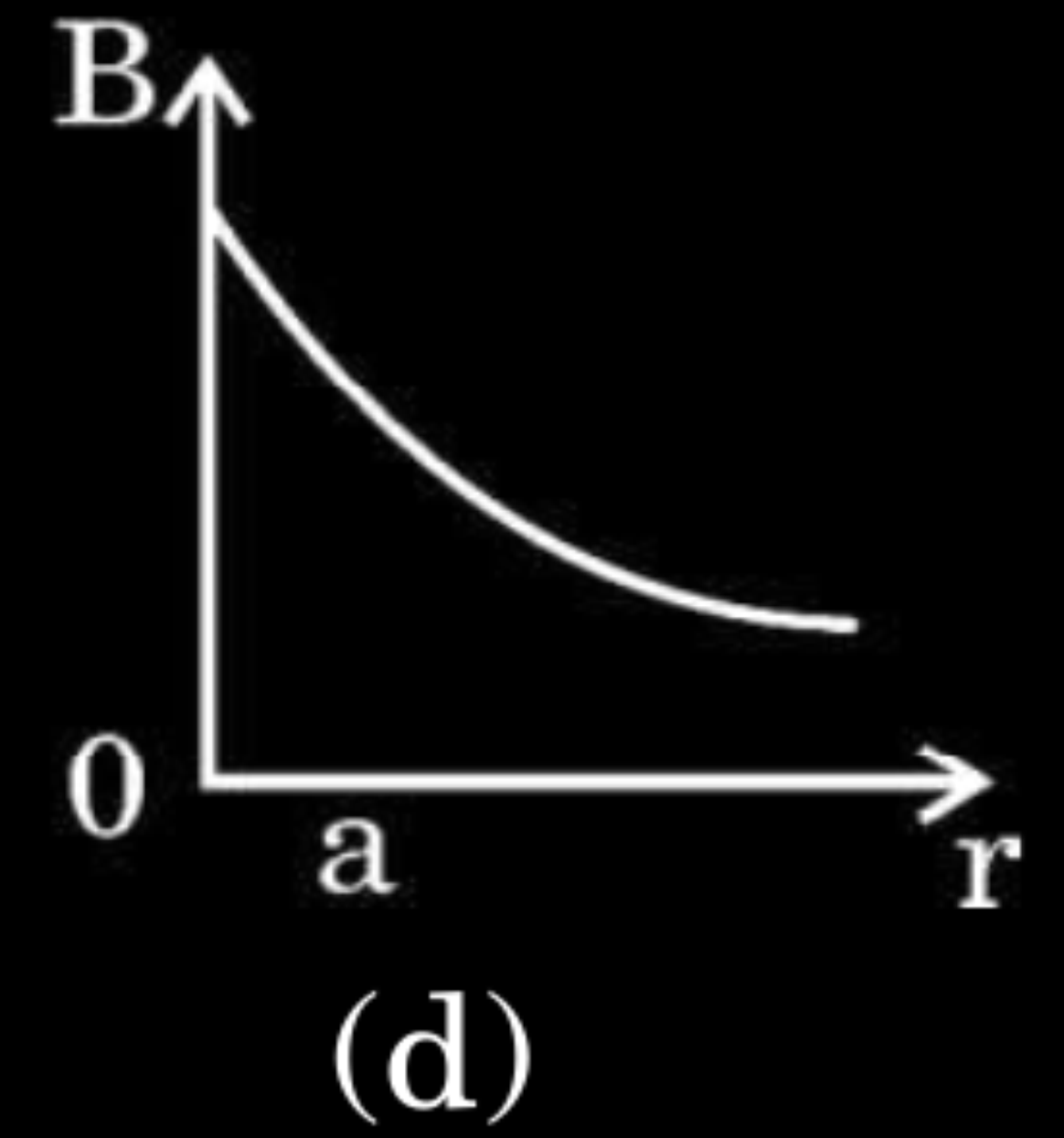
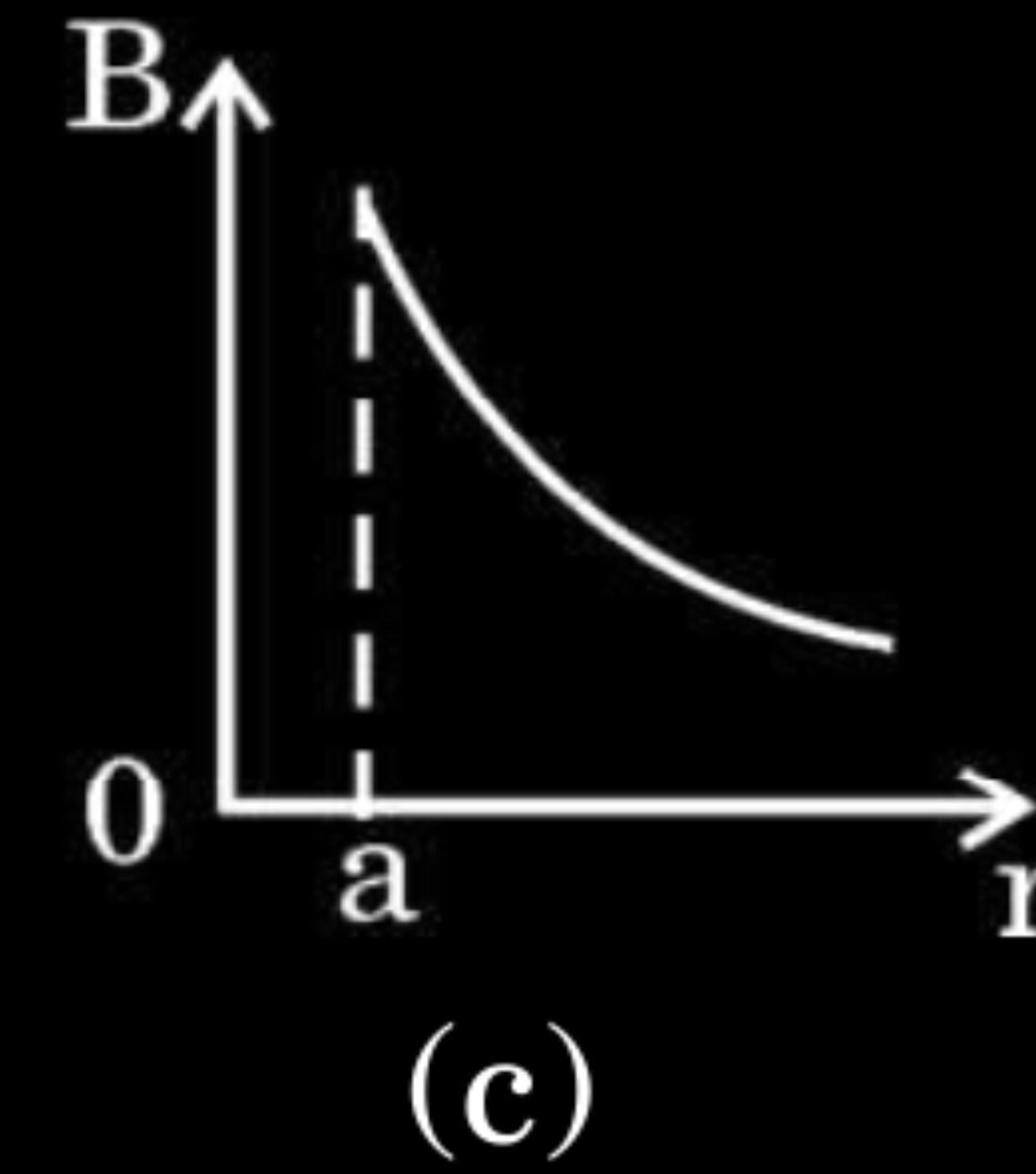
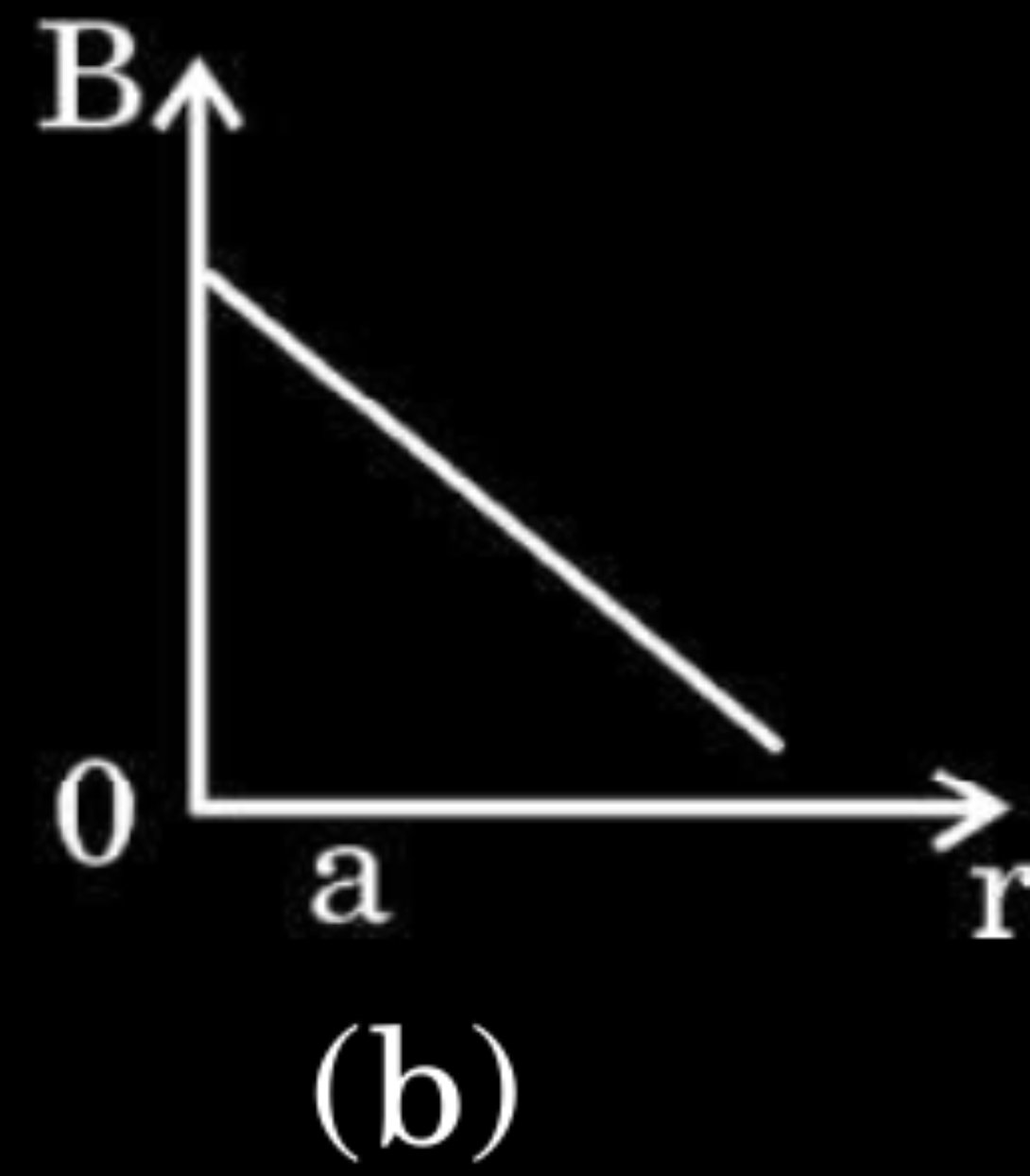
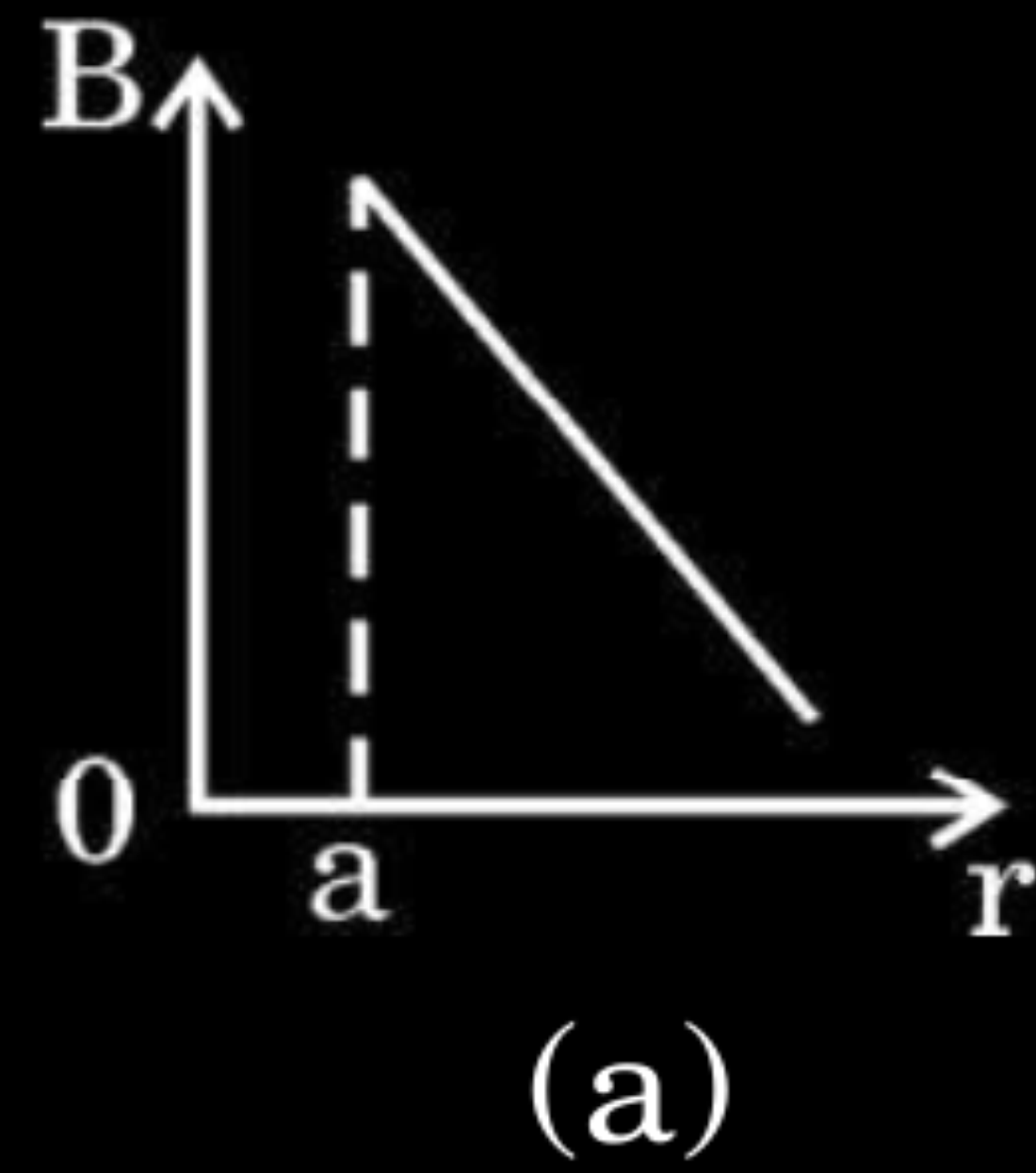
(a) Calculation of magnetic field due to current in A and B	$\frac{1}{2} + \frac{1}{2}$
Net magnetic field	1
(b) Force per unit length	1

(a) $B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{2 \times 10^{-7} \times 6}{1 \times 10^{-2}}$ $= 1.2 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2}$
$B_2 = \frac{\mu_0 I_2}{2\pi r} = \frac{2 \times 10^{-7} \times 2}{1 \times 10^{-2}}$ $= 0.4 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2}$
$\therefore B_1$ and B_2 are in same direction.	
$B_{net} = B_1 + B_2$ $= 1.2 \times 10^{-4} \text{ T} + 0.4 \times 10^{-4} \text{ T}$	
$= 1.6 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2} + \frac{1}{2}$
(b) Magnitude of magnetic force per unit length	
$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$	$\frac{1}{2}$
$= \frac{2 \times 10^{-7} \times 6 \times 2}{2 \times 10^{-2}} = 1.2 \times 10^{-4} \text{ N/m}$	$\frac{1}{2}$

15. Which of the following graphs correctly represents the variation of the magnitude of the magnetic field outside a straight infinite current carrying wire of radius 'a', as a function of distance 'r' from the centre of the wire ?



15. Which of the following graphs correctly represents the variation of the magnitude of the magnetic field outside a straight infinite current carrying wire of radius 'a', as a function of distance 'r' from the centre of the wire ?



1. A long straight wire of radius 'a' carries a steady current 'I'. The current is uniformly distributed across its area of cross-section. The ratio of magnitude of magnetic field $\vec{B}_1$ at $\frac{a}{2}$ and $\vec{B}_2$ at distance 2a is

1

(a) $\frac{1}{2}$

(b) 1

(c) 2

(d) 4

1. A long straight wire of radius 'a' carries a steady current 'I'. The current is uniformly distributed across its area of cross-section. The ratio of magnitude of magnetic field $\vec{B}_1$ at $\frac{a}{2}$ and $\vec{B}_2$ at distance 2a is

1

(a) $\frac{1}{2}$

(b) 1

(c) 2

(d) 4

2. Two horizontal thin long parallel wires, separated by a distance r carry current I each in the opposite directions. The net magnetic field at a point midway between them, will be

1

- (A) zero
- (B) $\left(\frac{\mu_0 I}{2\pi r}\right)$ vertically downward
- (C) $\left(\frac{2\mu_0 I}{r}\right)$ vertically upward
- (D) $\left(\frac{\mu_0 I}{\pi r}\right)$ vertically downward

2. Two horizontal thin long parallel wires, separated by a distance r carry current I each in the opposite directions. The net magnetic field at a point midway between them, will be

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- (D) $\left(\frac{\mu_0 I}{\pi r}\right)$ vertically downward

No option is correct.

[Award one mark to each student]

- 6.** A long solenoid of length L and number of turns N carrying a current I produces a magnetic field B along its axis. If the current in the solenoid is doubled and its number of turns halved, the magnetic field along its axis will be :

(a) $2B$

(b) $4B$

(c) $\frac{B}{2}$

(d) B

6. A long solenoid of length L and number of turns N carrying a current I produces a magnetic field B along its axis. If the current in the solenoid is doubled and its number of turns halved, the magnetic field along its axis will be :

(a) $2B$

(b) $4B$

(c) $\frac{B}{2}$

(d) B

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- (b) (i) Derive an expression for the force acting on a conductor carrying current I in a magnetic field $\vec{B}$.
- (ii) A long horizontal conductor carries a current of 30 A, in east to west direction. What are the magnitude and direction of the magnetic field due to the current 1.0 m below the conductor ?

5

(i) Deriving expression for force on a current carrying wire

3

(ii) Finding the magnitude and direction of magnetic field

1+1

(i) Consider a wire of length l , area of cross- section A and having number density of free electrons n and carrying electric current I . Total number of charge carriers in the wire = nIA . If $\vec{v}_d$ is drift velocity of electrons, magnetic force experience by the electrons is $\vec{F} = -(nIA)e\vec{v}_d \times \vec{B}$ As $I\vec{l} = -enIA\vec{v}_d$ $\vec{F} = I(\vec{l} \times \vec{B})$	$\frac{1}{2}$
(ii) $B = \frac{2\mu_o I}{4\pi r}$	$\frac{1}{2}$
$B = \frac{2 \times 10^{-7} \times 30}{1} = 0.6 \times 10^{-5} T$	$\frac{1}{2}$
Magnetic field is directed towards SOUTH	1

17. *Assertion (A) :* The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason (R): The magnetic field inside the solenoid is uniform.

17. *Assertion (A) :* The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason (R): The magnetic field inside the solenoid is uniform.

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

22. (b) (i) State and explain Ampere's circuital law.
- (ii) Two long straight parallel wires separated by 20 cm, carry 5 A and 10 A current respectively, in the same direction. Find the magnitude and direction of the net magnetic field at a point midway between them.

(i) Statement and explanation of Ampere's circuital law	1
(ii) Finding magnitude and direction of magnetic field	2

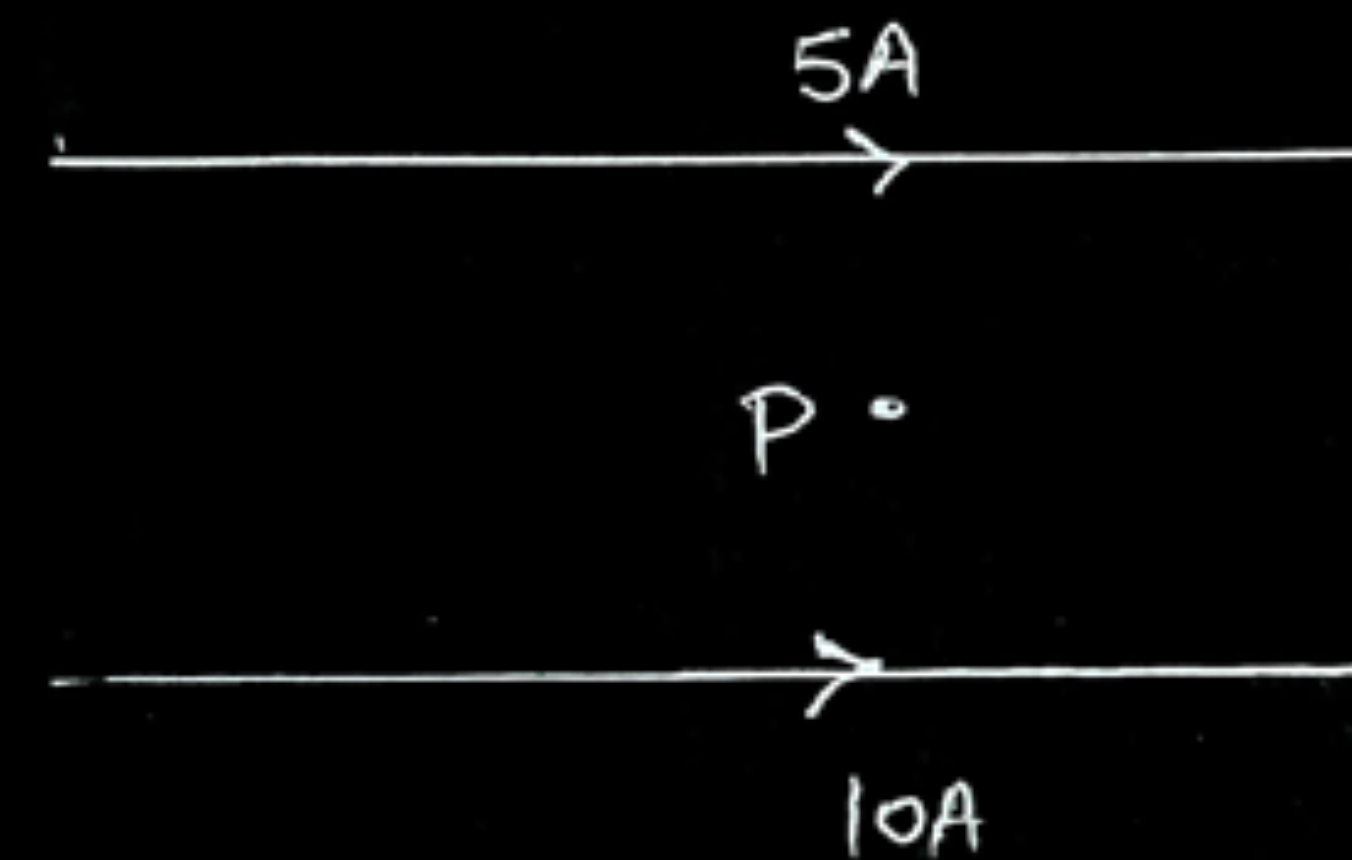
Line integral of magnetic field over a closed loop in vacuum is equal to μ_0 times the total current passing through the loop.

Alternatively

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

The integral in this expression is over a closed loop coinciding with the boundary of the surface.

(ii)



$$B = \frac{\mu_0 I}{2\pi r}$$

Net magnetic field $B = B_2 - B_1$

$$B = \frac{\mu_0 \times 10^2}{20\pi} [10 - 5]$$

$$B = \frac{4\pi \times 10^{-7} \times 10^2 \times 5}{20\pi}$$

$$B = 10^{-5} T$$

Along the direction of magnetic field produced by the conductor carrying current 10A.

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

5. A straight wire is kept horizontally along east-west direction. If a steady current flows in wire from east to west, the magnetic field at a point above the wire will point towards

(A) East

(B) West

(C) North

(D) South

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1

(A) East

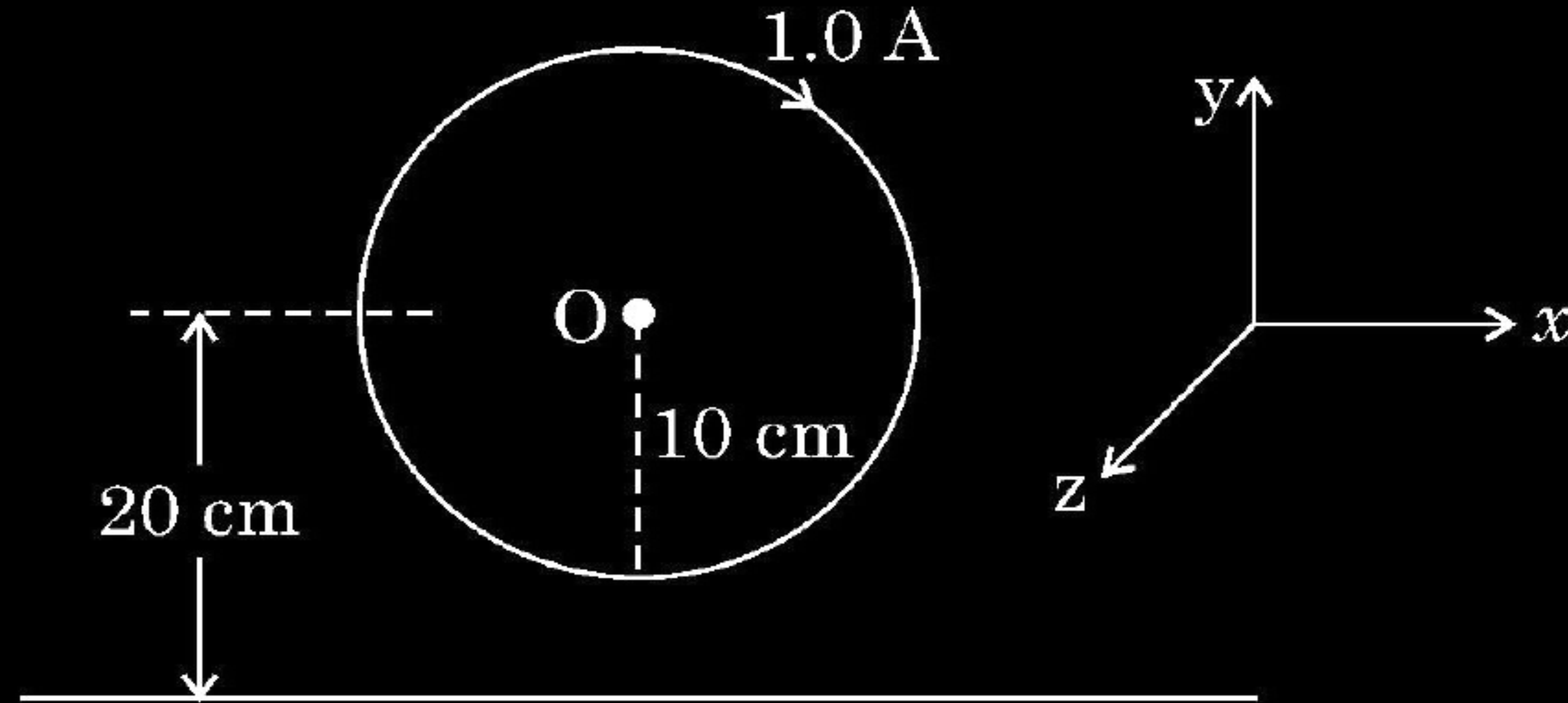
(B) West

(C) North

(D) South

(C) North

26. A circular loop of radius 10 cm carrying current of 1.0 A lies in x - y plane. A long straight wire lies in the same plane parallel to x -axis at a distance of 20 cm as shown in figure.



Find the direction and value of current that has to be maintained in the wire so that the net magnetic field at O is zero.

3

Finding value of current in a long straight wire	$2 \frac{1}{2}$
Finding direction of current in a long straight wire	$\frac{1}{2}$

Magnetic field due to circular current loop at its centre O.

$$B_1 = \frac{\mu_0 I_1}{2r}$$

$$= \frac{\mu_0 \times 1}{2 \times 0.1}$$

$$= 5\mu_0 \text{ T}$$

The magnetic field B_1 is perpendicular to plane of loop and directed inwards.

Magnetic field due to long current carrying straight wire at O.

$$B_2 = \frac{\mu_0 I_2}{2\pi r}$$

$$B_2 = \frac{\mu_0 I_2}{2\pi \times 0.2} \text{ T}$$

For net magnetic field at O to be zero, B_1 should be equal and opposite to B_2 .

$$5\mu_0 = \frac{\mu_0 I_2}{0.4\pi}$$

$$I_2 = 2\pi \text{ A}$$

$$= 6.28 \text{ A}$$

Direction of current in the straight wire is along +ve x axis.

Alternatively

Net magnetic field at O is zero.

$$B_{\text{loop}} = B_{\text{wire}}$$

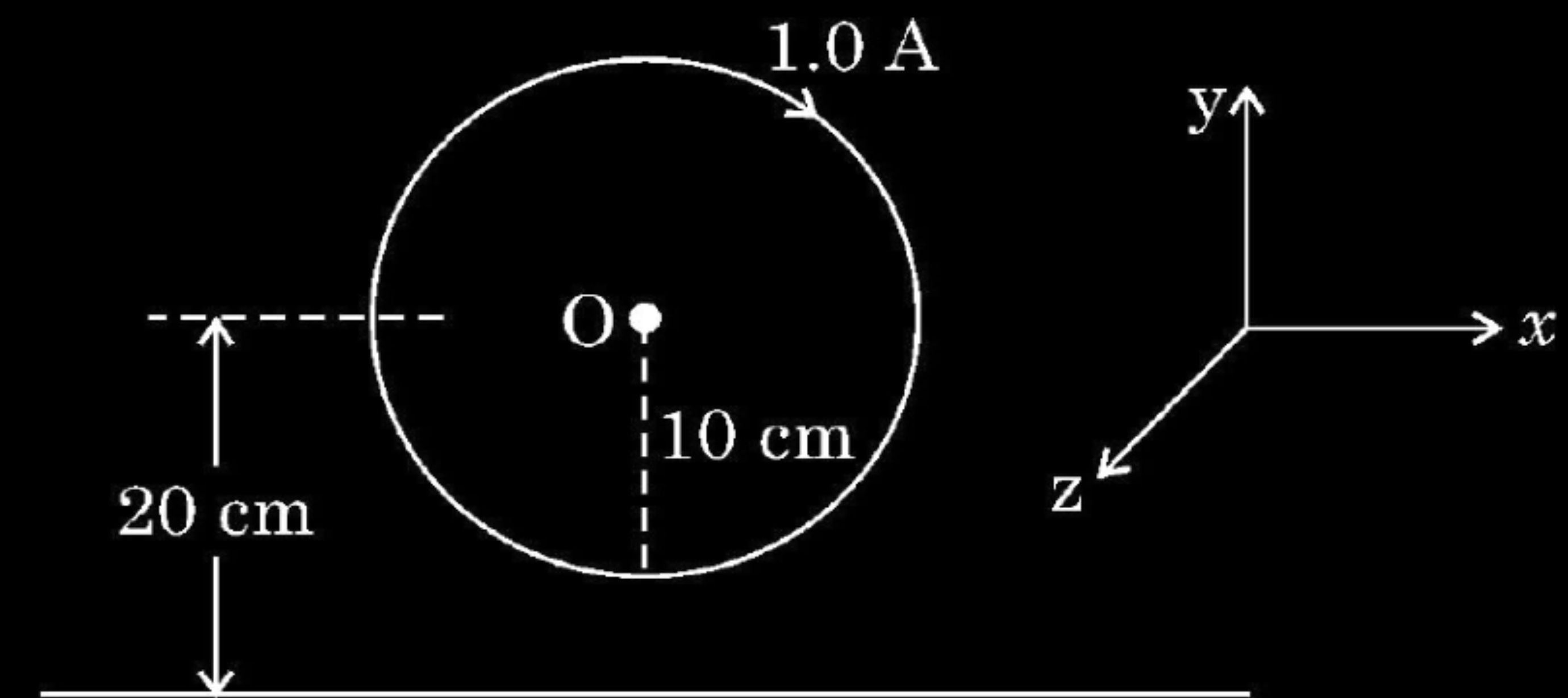
$$\frac{\mu_0 I_1}{2r_1} = \frac{\mu_0 I_2}{2\pi r_2}$$

$$\frac{\mu_0 \times 1}{2 \times 0.1} = \frac{\mu_0 I_2}{2\pi \times 0.2}$$

$$I_2 = 2\pi \text{ A}$$

$$= 6.28 \text{ A}$$

Direction of current is along + x-axis.



1/2

1/2

1/2

1/2

1/2

1/2

1/2

1

1/2

1/2

1/2

4. The magnetic field B within the solenoid of length L having N turns and carrying current I will be :

(A) $\mu_0 N I L$ (B) $\frac{\mu_0 N I}{L}$

(C) $\frac{\mu_0 N I}{4\pi L}$ (D) $\mu_0 N I$

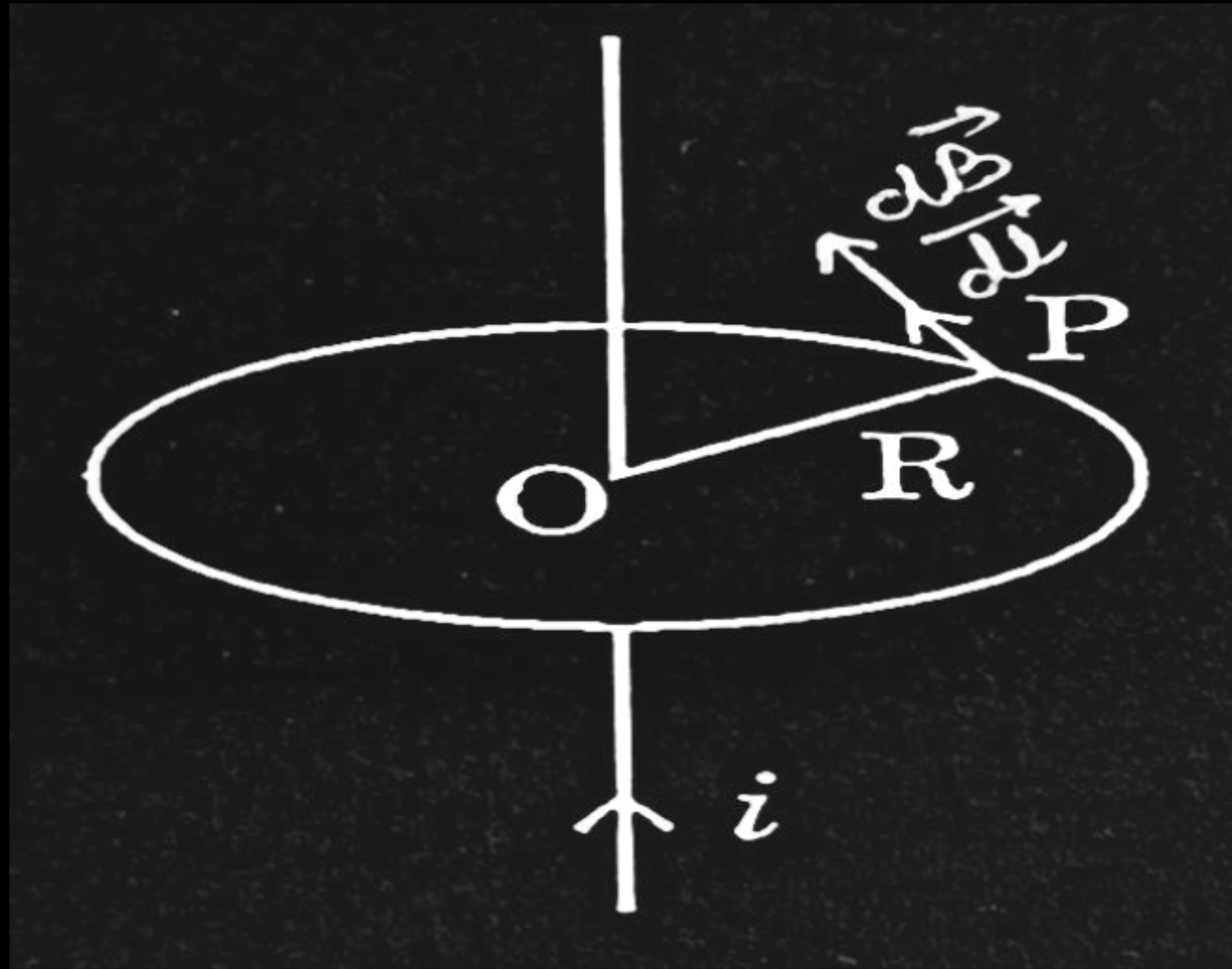
(B) $\frac{\mu_0 N I}{L}$

- 27.** (a) Use Ampere's law to derive the expression for the magnetic field due to a long straight current carrying wire of infinite length.
- (b) Why is Ampere's law used for the derivation in (a) above and not Biot-Savart's law ? Explain.

3

(a) Deriving expression of magnetic field for long straight wire	2
(b) Explanation	1

(a)



$\frac{1}{2}$

Consider an amperian loop of radius R around the wire carrying current I .

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$\frac{1}{2}$

$$B \times 2\pi r = \mu_0 i$$

$\frac{1}{2}$

$$B = \frac{\mu_0 i}{2\pi r}$$

$\frac{1}{2}$

(b) The expression for magnetic field due to a long straight current carrying infinite wire can be obtained easily using Ampere's circuital law whereas Biot- Savart's law requires lengthy calculation.

1

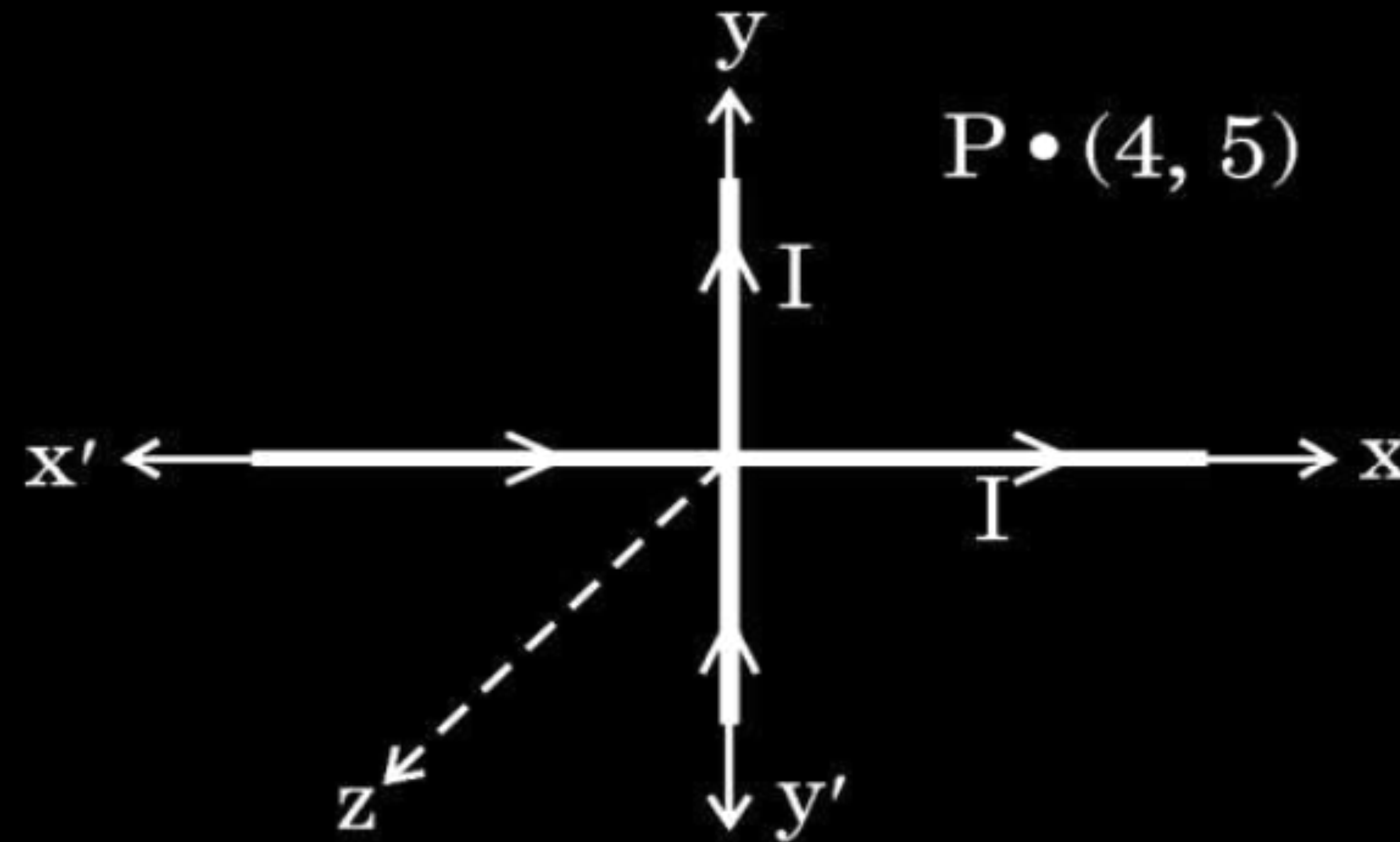
8. A long straight wire is held vertically and carries a steady current in upward direction. The shape of magnetic field lines produced by the current-carrying wire are :
- (A) horizontal straight lines directed radially out from the wire.
 - (B) straight lines parallel to the current-carrying wire.
 - (C) concentric horizontal circles around the wire.
 - (D) coaxial helixes around the wire.

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(C) Concentric horizontal circles around the wire

- 23.** (a) Write vector form of Biot-Savart law.
- (b) Two insulated long straight wires, each carrying 2.0 A current are kept along xx' and yy' axis as shown in the figure. Find the magnitude and direction of resultant magnetic field at point P (4m, 5m).

3



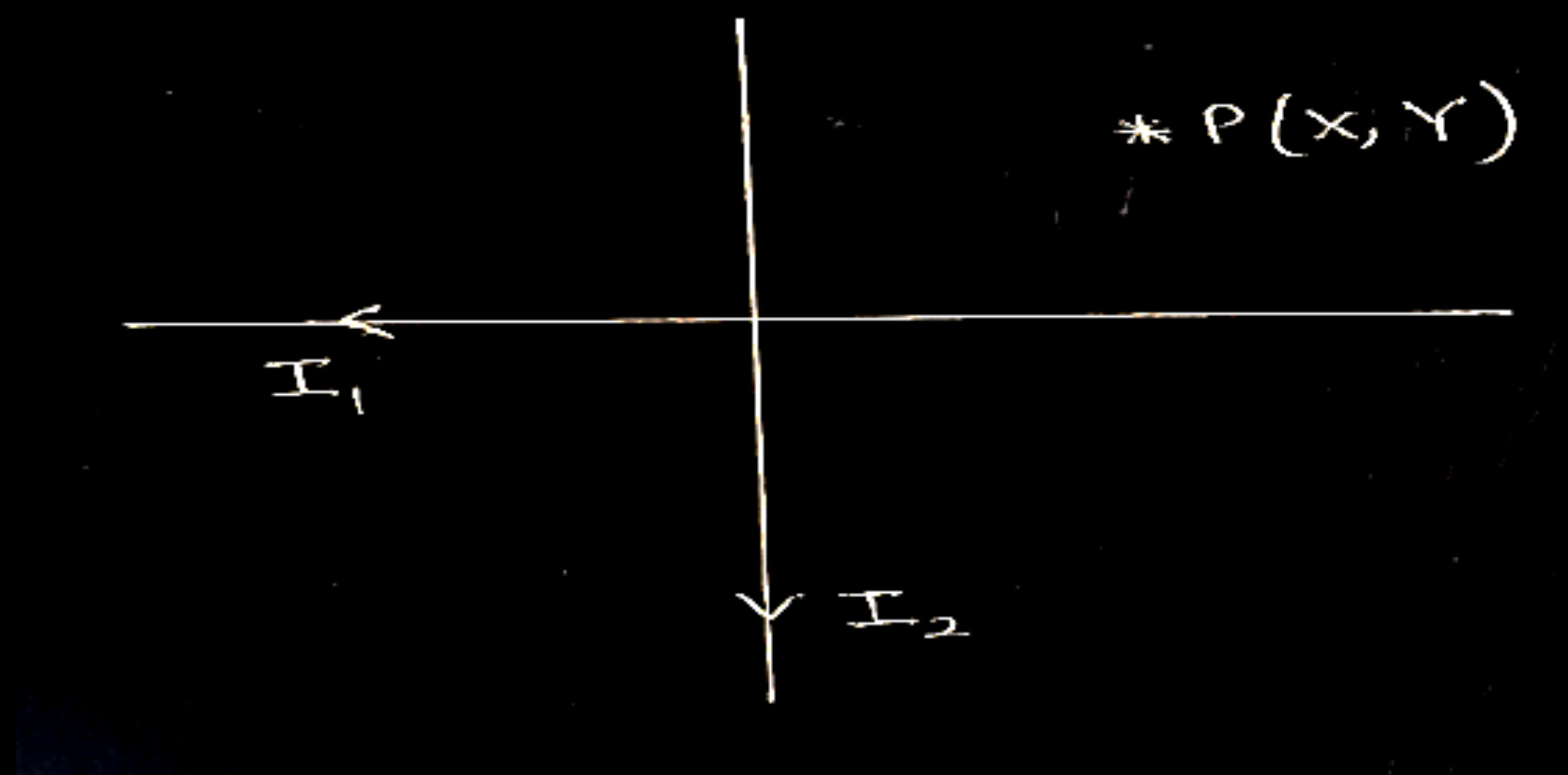
(a) Vector form of Biot – Savart Law	1
(b) Finding ,	
• Magnitude of resultant magnetic field	$1\frac{1}{2}$
• Direction of resultant magnetic field	$\frac{1}{2}$

(a) $d\vec{B} = \frac{\mu_o}{4\pi} \frac{I(\vec{dl} \times \vec{r})}{r^3}$	1
(b) As $B = \frac{\mu_o}{4\pi} \frac{2I}{a}$	
$\therefore B_1 = \text{Magnetic field due to current carrying wire along } XX'$	
$\vec{B}_1 = \frac{\mu_o}{4\pi} \frac{2I}{a} = \frac{2 \times 2 \times 10^{-7}}{5} = (8 \times 10^{-8} \text{ T})(\hat{k})$	$\frac{1}{2}$
$B_2 = \text{Magnetic field due to current carrying wire along } YY'$	
$\vec{B}_2 = \frac{\mu_o}{4\pi} \frac{2I}{a} = \frac{2 \times 2 \times 10^{-7}}{4} = 10 \times 10^{-8} \text{ T } (-\hat{k})$	$\frac{1}{2}$
$\vec{B}_{\text{net}} = \vec{B}_2 + \vec{B}_1$	$\frac{1}{2}$
$\quad = 2 \times 10^{-8} \text{ T } (-\hat{k})$	
• Direction of resultant magnetic field along $-Z$ axis.	$\frac{1}{2}$

- (b) Two infinitely long conductors kept along XX' and YY' axes are carrying current I_1 and I_2 along $-X$ axis and $-Y$ axis respectively. Find the magnitude and direction of the net magnetic field produced at point $P(X, Y)$.

Finding the magnitude and direction of the net magnetic field

2+1



Magnetic field due to conductor carrying current I_1 ($\vec{B}_1$) = $\frac{\mu_0 I_1}{2\pi Y} (-\hat{k})$

$\frac{1}{2}$

Magnetic field due to conductor Carrying current I_2 ($\vec{B}_2$) = $\frac{\mu_0 I_2}{2\pi X} (\hat{k})$

$\frac{1}{2}$

$$\vec{B}_p = \vec{B}_1 + \vec{B}_2$$

$$\vec{B}_p = \frac{\mu_0}{2\pi} \left[\frac{I_2}{X} - \frac{I_1}{Y} \right] \hat{k}$$

1

1

Direction will be along the Z-axis.

3. A current I flows through a long straight conductor which is bent into a circular loop of radius R in the middle as shown in the figure.



The magnitude of the net magnetic field at point O will be

- (a) Zero (b) $\frac{\mu_0 I}{2R} (1 + \pi)$ (c) $\frac{\mu_0 I}{4\pi R}$ (d) $\frac{\mu_0 I}{2R} \left(1 - \frac{1}{\pi}\right)$ **1**

3. A current I flows through a long straight conductor which is bent into a circular loop of radius R in the middle as shown in the figure.



The magnitude of the net magnetic field at point O will be

- (a) Zero (b) $\frac{\mu_0 I}{2R} (1 + \pi)$ (c) $\frac{\mu_0 I}{4\pi R}$ (d) $\frac{\mu_0 I}{2R} \left(1 - \frac{1}{\pi}\right)$ **1**

(d) $\frac{\mu_0 I}{2R} \times \left(1 - \frac{1}{\pi}\right)$

7. A current of 10A is flowing from east to west in a long straight wire kept on a horizontal table. The magnetic field developed at a distance of 10 cm due north on the table is :

- (a) 2×10^{-5} T, acting downwards (b) 2×10^{-5} T, acting upwards
(c) 4×10^{-5} T, acting downwards (d) 4×10^{-5} T, acting upwards

7. A current of 10A is flowing from east to west in a long straight wire kept on a horizontal table. The magnetic field developed at a distance of 10 cm due north on the table is :

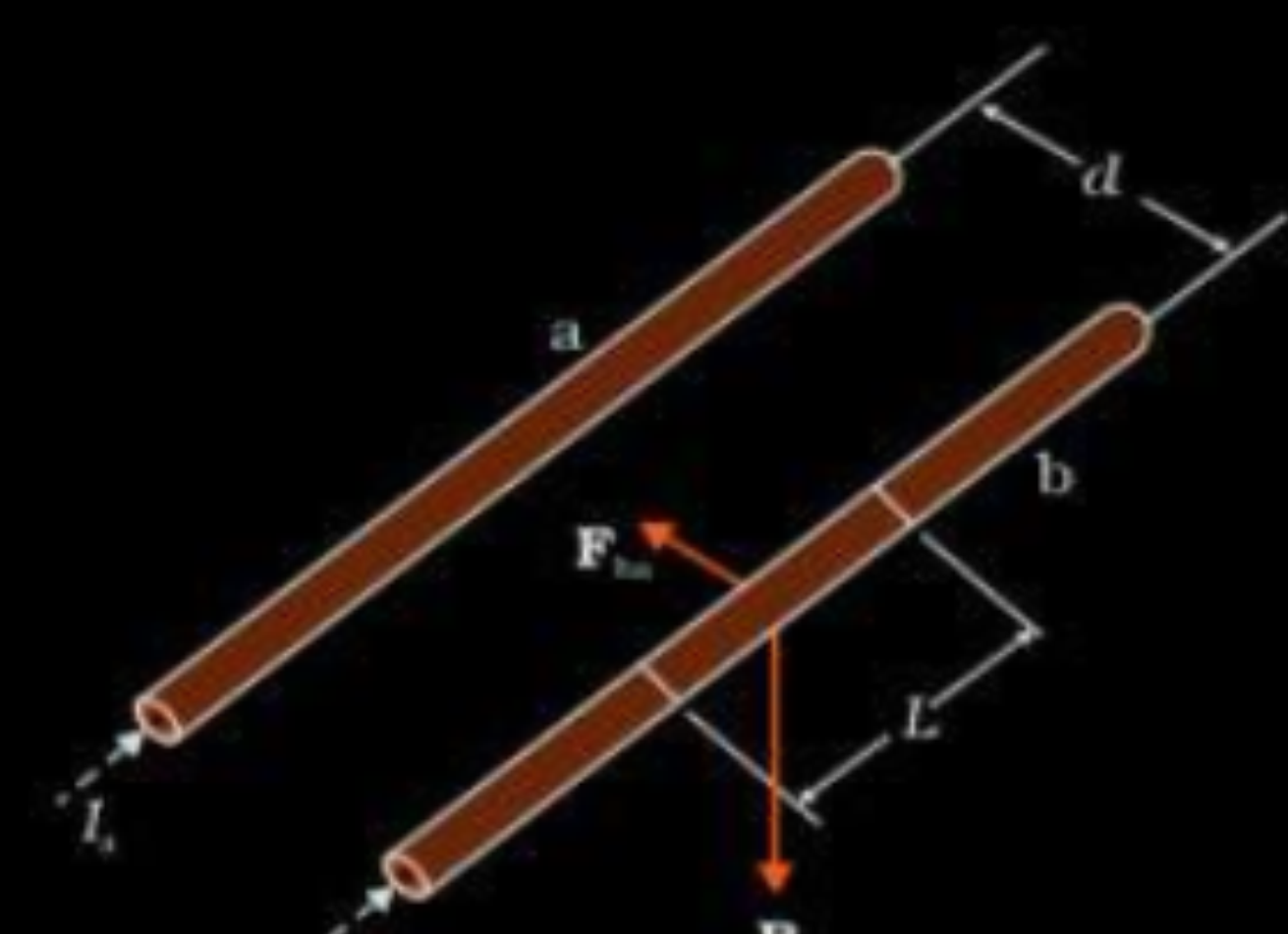
- (a) 2×10^{-5} T, acting downwards (b) 2×10^{-5} T, acting upwards
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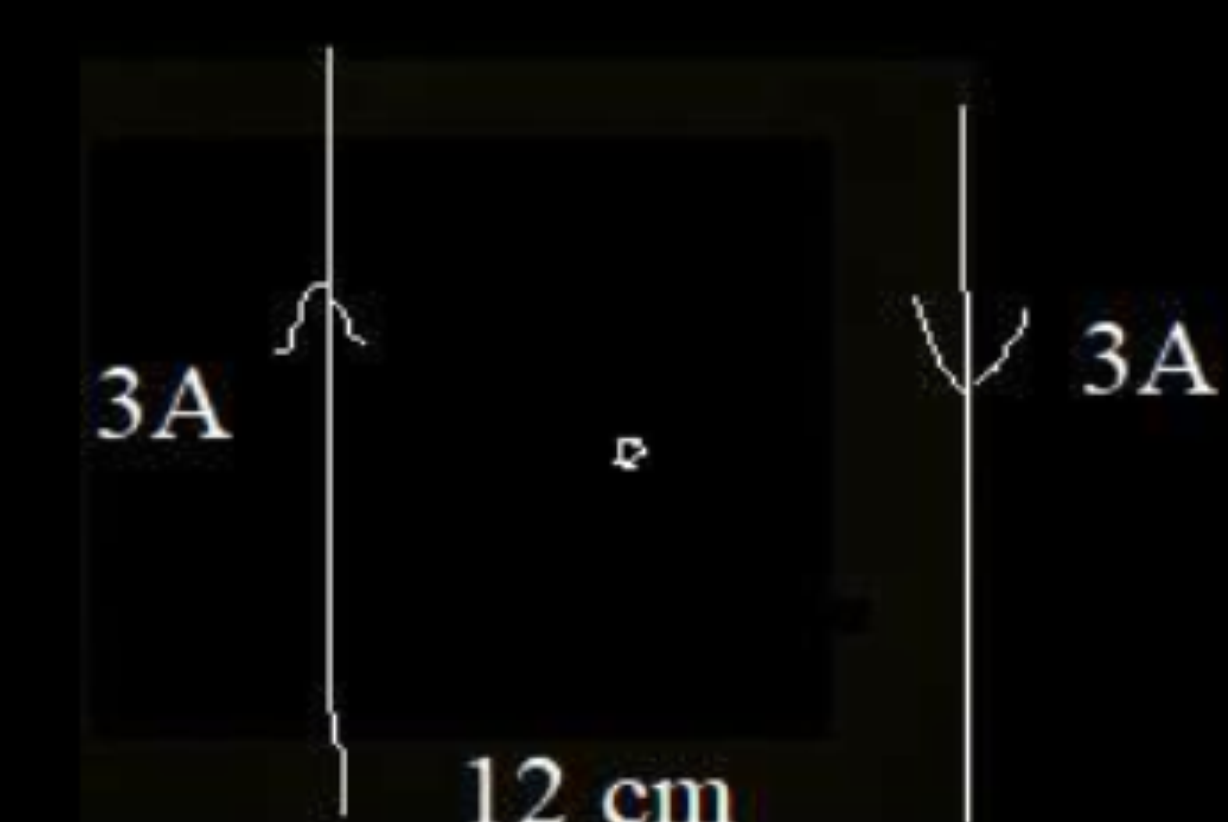
1

(a) 2×10^{-5} T acting downwards

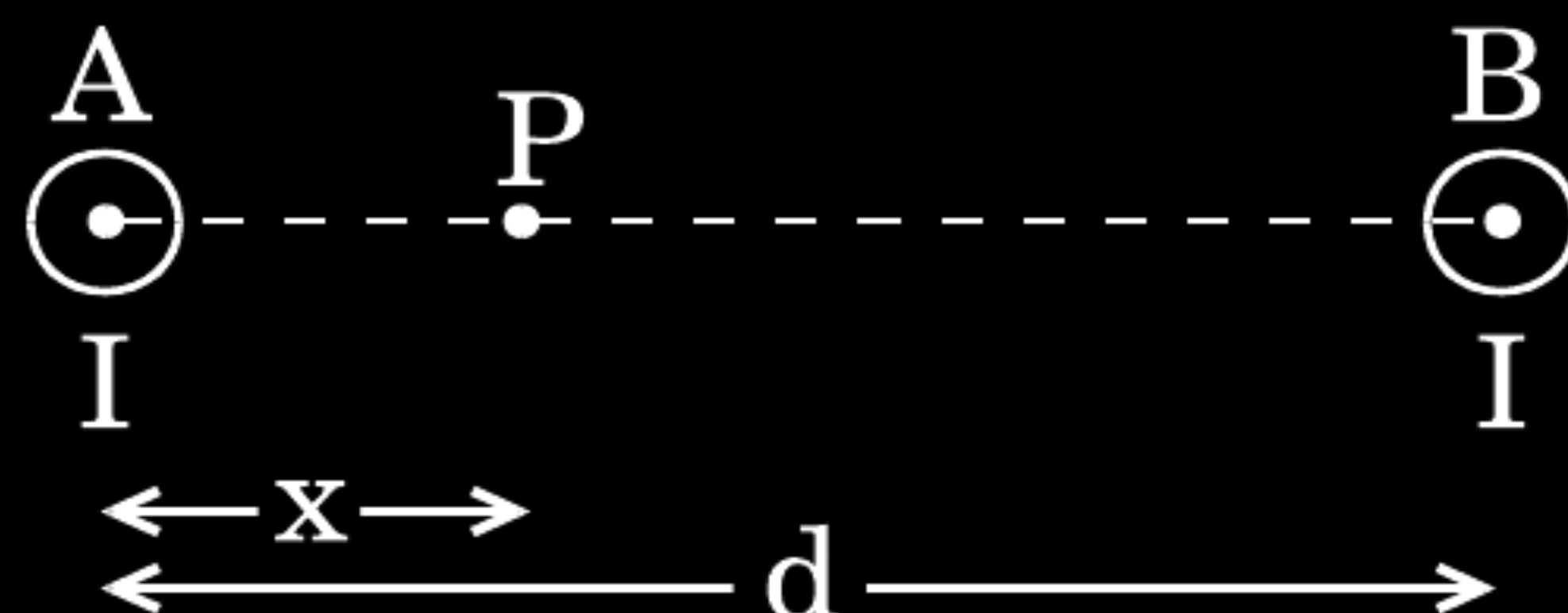
- 36.** (a) Derive the expression for the force acting per unit length between two long straight parallel current carrying conductors. Hence define one ampere.
- (b) Two long parallel straight conductors are placed 12 cm apart in air. They carry equal currents of 3 A each. Find the magnitude and direction of the magnetic field at a point midway between them (drawing a figure) when the currents in them flow in opposite directions.

a) Diagram showing direction	1/2 mark
Derivation of expression	1 1/2 marks
Definition of 1 ampere	1 mark
b) Magnetic fields of the wires	1/2 mark
Net magnetic field and its direction	1 + 1/2 mark

(a) 	
Magnetic field due to the current I_a flowing in conductor 'a' at any point on conductor 'b' $B_a = \frac{\mu_0 I_a}{2\pi d}$	1/2
(Acting perpendicular downward) Therefore force on conductor 'b' due to field B_a	
$\vec{F} = I_b (\vec{l}_b \times \vec{B}_a)$ $ \vec{F}_{ba} = I_b l_b \times \frac{\mu_0 I_a}{2\pi d}$ $= \frac{\mu_0 I_a I_b l_b}{2\pi d}$ $\frac{ \vec{F}_{ba} }{l_b} = \frac{\mu_0 I_a I_b}{2\pi d}$	1/2
Definition of 1 A : Two straight infinitely long parallel conductors are said to carry 1 A current each when they interact each other with a force of $2 \times 10^{-7} \text{ Nm}^{-1}$, when kept 1m apart in vacuum	1

	
$\vec{B} = \vec{B}_1 + \vec{B}_2$	1/2
$B = \frac{\mu_0 I_1}{2\pi r_1} + \frac{\mu_0 I_2}{2\pi r_2}$	
$= \frac{\mu_0}{2\pi} \left(\frac{3}{6 \times 10^{-2}} + \frac{3}{6 \times 10^{-2}} \right)$	
$= \frac{4\pi \times 10^{-7} \times 3}{\pi \times 6 \times 10^{-2}}$	
$= 2 \times 10^{-5} \text{ tesla}$	1
Direction of $\vec{B}$ at midpoint is perpendicular to the plane containing the two conductors and pointing downwards. (Note: give full credit of this direction if student takes direction opposite to the shown in fig and answer accordingly)	1/2

25. Two long straight parallel wires A and B separated by a distance d , carry equal current I flowing in same direction as shown in the figure.



- (a) Find the magnetic field at a point P situated between them at a distance x from one wire.
- (b) Show graphically the variation of the magnetic field with distance x for $0 < x < d$.

2

Magnetic field at point P

1 ½ mark

Curve

½ mark

a)

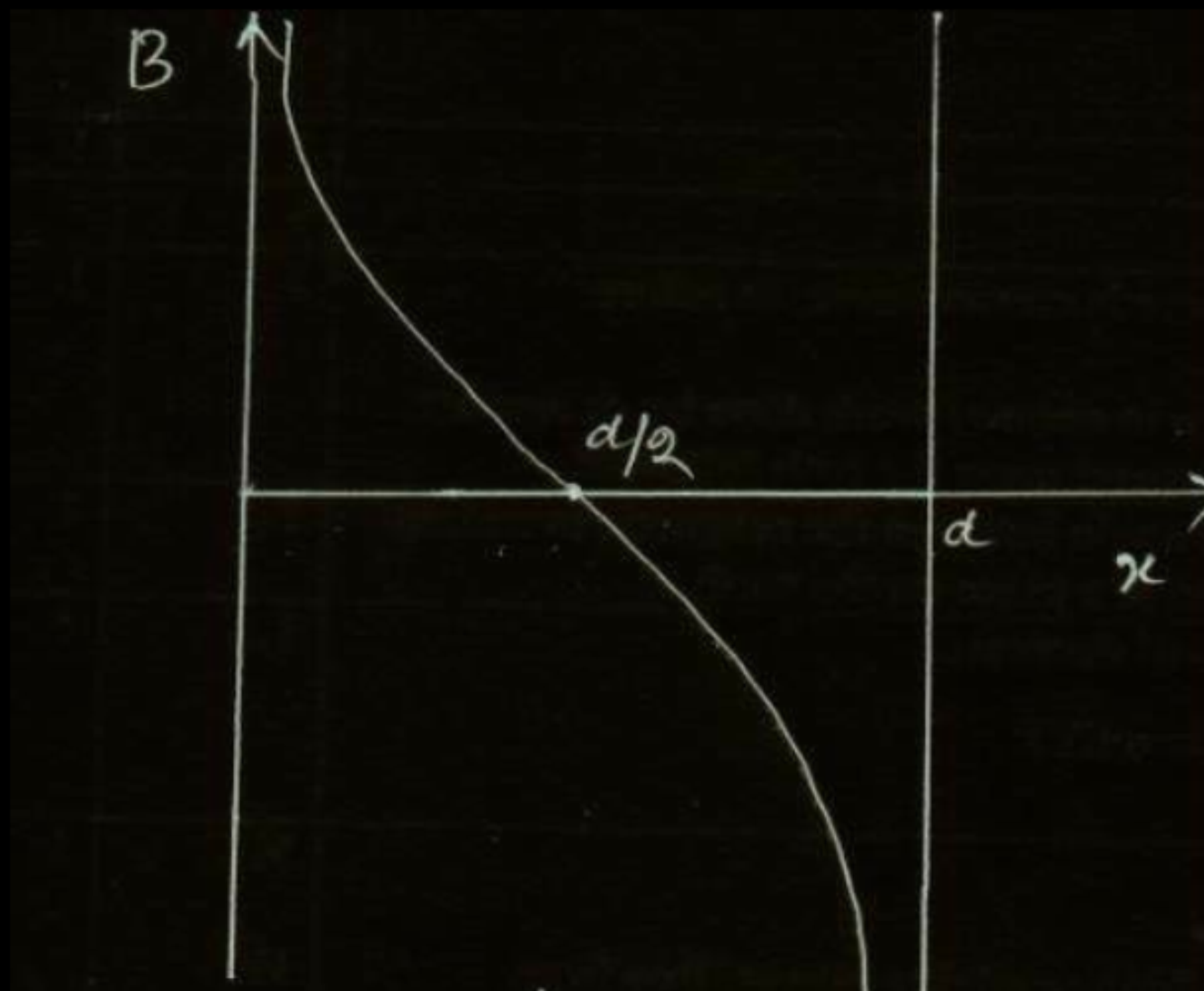
$$B = \frac{\mu_0 I}{2\pi x}$$

$$B_P = B_1 - B_2 = \frac{\mu_0 I}{2\pi x} - \frac{\mu_0 I}{2\pi(d-x)} = \frac{\mu_0 I(d-2x)}{2\pi(d-x)x}$$

 $\frac{1}{2}$

1

b)

 $\frac{1}{2}$

State Ampere's circuital law. Use this law to find magnetic field due to straight infinite current carrying wire. How are the magnetic field lines different from the electrostatic field lines ?

Statement of Ampere's Circuital law	1
Finding Magnetic Field	1 ½
Differences between the two types of field lines	½

According to Ampers's circuital law, "The line integral of the magnetic field, around a closed loop, equals μ_o times the total current passing through the surface enclosed by that loop."

Alternatively,

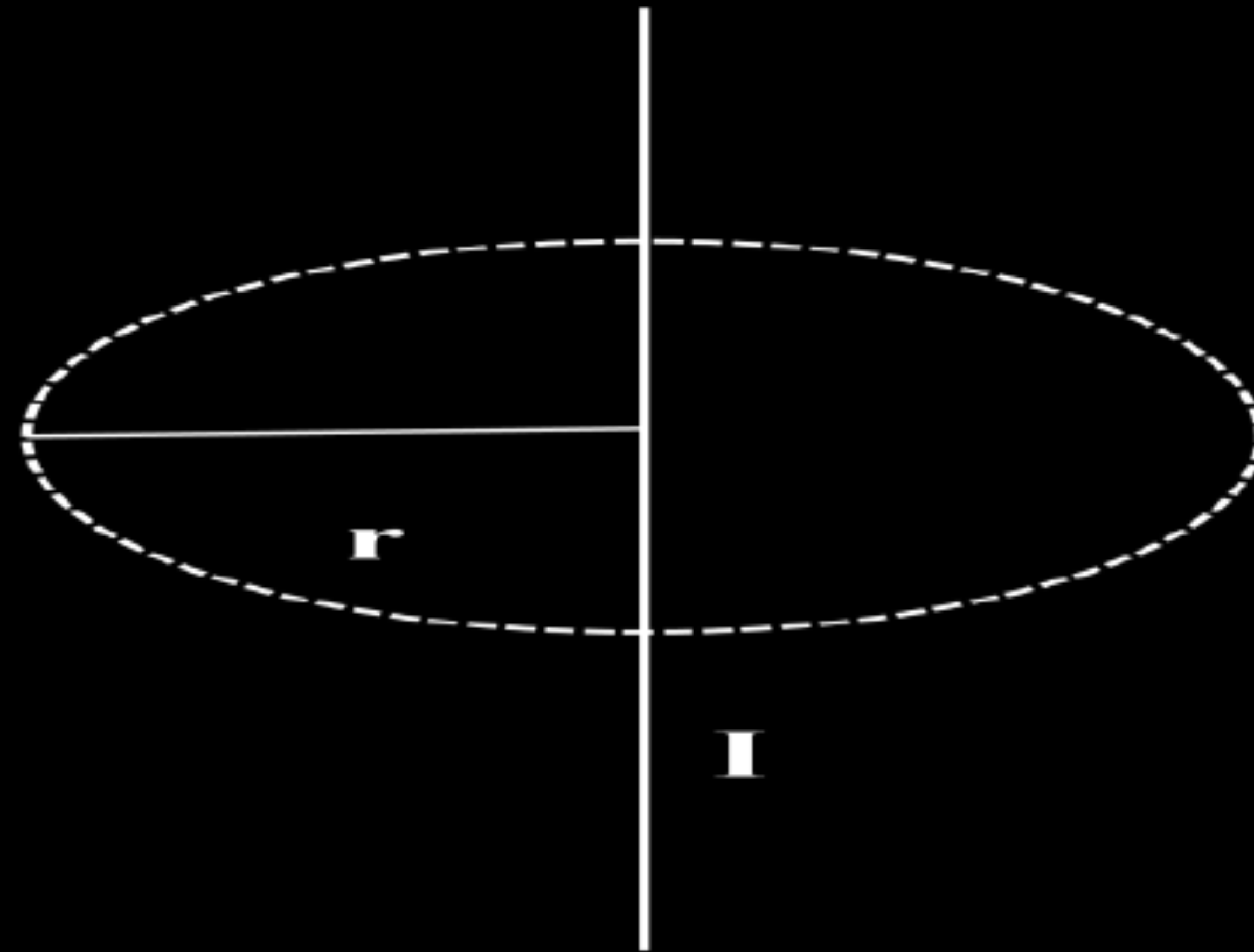
$$\oint \vec{B} \cdot d\vec{l} = \mu_o I$$

For the infinite current carrying wire, we get

$$\oint B \cdot dl = \mu_o I$$

$$\text{or } B 2\pi r = \mu_o I$$

$$\text{or } B = \frac{\mu_o I}{2\pi r}$$



The magnetic field lines form closed loops while the electric field lines originate from positive charges and end at negative charges.

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$